MPG No. 3.3.8 2′ [Type III, monoclinic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$
$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$
$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$
$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & T_{zzz} \\ 0 & T_{yyz} & 0 \\ T_{xxz} & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$
$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$
$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$
$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & 0 \\ 0 & -\frac{T_{yyz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$
\begin{aligned}
T_{xxz} &= 2\mathbb{M}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{z} \\
T_{yyx} &= 2\mathbb{M}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{x} \\
T_{yyz} &= -2\mathbb{M}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{z} \\
T_{zzx} &= -2\mathbb{M}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{x} \\
\frac{T_{yyz}}{2} &= -\mathbb{M}^{(1)}_{xy} + \mathbb{T}^{(2)}_{z} \\
\frac{T_{xxz}}{2} &= \mathbb{M}^{(1)}_{xy} + \mathbb{T}^{(2)}_{z} \\
\frac{T_{zzx}}{2} &= -\mathbb{M}^{(1)}_{yz} + \mathbb{T}^{(2)}_{x} \\
\frac{T_{yyx}}{2} &= \mathbb{M}^{(1)}_{yz} + \mathbb{T}^{(2)}_{x}
\end{aligned}
$$

* Rank 3 tensor (a).

$$
\begin{bmatrix}
0 & T_{yzy} & 0 \\
T_{zxx} & 0 & T_{zxz} \\
0 & T_{xyy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzy} &= \mathbb{M}^{(2)}_{xy} + \mathbb{T}^{(3)}_{z} \\
T_{zxx} &= \mathbb{M}^{(2)}_{xy} - \mathbb{T}^{(3)}_{z} \\
T_{zxz} &= \mathbb{M}^{(2)}_{yz} + \mathbb{T}^{(3)}_{x} \\
T_{xyy} &= \mathbb{M}^{(2)}_{yz} - \mathbb{T}^{(3)}_{x}
\end{aligned}
$$

* Rank 4 tensor (sss).

$$
\begin{bmatrix}
0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} \\
0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} \\
0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} \\
T_{xxyz} & T_{yyyz} & T_{zzyz} & 0 & T_{yzzx} & 0 \\
0 & 0 & 0 & T_{yzzx} & 0 & T_{zxxy} \\
T_{xxxy} & T_{yyxy} & T_{zzxy} & 0 & T_{zxxy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxyz} &= 2\mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz} \\
T_{xxxy} &= \mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} \\
T_{yyyz} &= \mathbb{T}^{\alpha(1)}_{4x} - \mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{yz} \\
T_{yyxy} &= -\mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} \\
T_{zzyz} &= -\mathbb{T}^{\alpha(1)}_{4x} - \mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{yz} \\
T_{zzxy} &= 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} \\
T_{yzzx} &= 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy} \\
T_{zxxy} &= 2\mathbb{T}^{\beta(1)}_{4x} + \mathbb{T}^{(2)}_{yz}
\end{aligned}
$$

* Rank 4 tensor (ssa).

$$
\begin{bmatrix}
0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} \\
0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} \\
0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} \\
-T_{xxyz} & -T_{yyyz} & -T_{zzyz} & 0 & T_{yzzx} & 0 \\
0 & 0 & 0 & -T_{yzzx} & 0 & T_{zxxy} \\
-T_{xxxy} & -T_{yyxy} & -T_{zzxy} & 0 & -T_{zxxy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxyz} &= -4\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{xxxy} &= 2\mathbb{M}_{z}^{(1)} - 2\mathbb{M}_{z}^{\alpha(1)} + 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yyyz} &= 2\mathbb{M}_{x}^{(1)} - 2\mathbb{M}_{x}^{\alpha(1)} + 2\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yyxy} &= -2\mathbb{M}_{z}^{(1)} + 2\mathbb{M}_{z}^{\alpha(1)} + 2\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{zzyz} &= -2\mathbb{M}_{x}^{(1)} + 2\mathbb{M}_{x}^{\alpha(1)} + 2\mathbb{M}_{x}^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{zzxy} &= -4\mathbb{M}_{z}^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yzzx} &= -\mathbb{M}_{z}^{(1)} - 4\mathbb{M}_{z}^{\alpha(1)} \\
T_{zxxy} &= -\mathbb{M}_{x}^{(1)} - 4\mathbb{M}_{x}^{\alpha(1)}
\end{aligned}
$$

* Rank 4 tensor (aas).

$$
\begin{bmatrix}
0 & T_{yzzx} & 0 \\
T_{yzzx} & 0 & T_{zxxy} \\
0 & T_{zxxy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzzx} &= 2\mathbb{T}_{xy}^{(4)} \\
T_{zxxy} &= 2\mathbb{T}_{yz}^{(4)}
\end{aligned}
$$

* Rank 4 tensor (aaa).

$$
\begin{bmatrix}
0 & T_{yzzx} & 0 \\
-T_{yzzx} & 0 & T_{zxxy} \\
0 & -T_{zxxy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzzx} &= -\mathbb{M}_{z}^{(2)} \\
T_{zxxy} &= -\mathbb{M}_{x}^{(2)}
\end{aligned}
$$

* Rank 4 tensor (sa).

$$
\begin{bmatrix}
T_{xxyz} & 0 & T_{xxxy} \\
T_{yyyz} & 0 & T_{yyxy} \\
T_{zzyz} & 0 & T_{zzxy} \\
0 & T_{yzzx} & 0 \\
T_{zxyz} & 0 & T_{zxxy} \\
0 & T_{xyzx} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxyz} &= \mathbb{M}_{x}^{(3)} + 2\mathbb{M}_{x}^{(4)} + 2\mathbb{M}_{x}^{\alpha(2)} \\
T_{xxxy} &= \mathbb{M}_{z}^{(3)} - \mathbb{M}_{z}^{\alpha(2)} + \mathbb{M}_{z}^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyz} &= \mathbb{M}_{x}^{(3)} - \mathbb{M}_{x}^{\alpha(2)} + \mathbb{M}_{x}^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyxy} &= \mathbb{M}_{z}^{(3)} - \mathbb{M}_{z}^{\alpha(2)} - \mathbb{M}_{z}^{\beta(2)} + 2\mathbb{T}_{xy}^{(5)} \\
T_{zzyz} &= \mathbb{M}_{x}^{(3)} - \mathbb{M}_{x}^{\alpha(2)} - \mathbb{M}_{x}^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzxy} &= \mathbb{M}_{z}^{(3)} + 2\mathbb{M}_{z}^{(4)} + 2\mathbb{M}_{z}^{\alpha(2)} \\
T_{yzzx} &= \mathbb{M}_{z}^{(4)} - \mathbb{M}_{z}^{\alpha(2)} - \mathbb{M}_{z}^{\beta(2)} - \mathbb{T}_{xy}^{(5)} \\
T_{zxyz} &= \mathbb{M}_{z}^{(4)} - \mathbb{M}_{z}^{\alpha(2)} + \mathbb{M}_{z}^{\beta(2)} + \mathbb{T}_{xy}^{(5)} \\
T_{zxxy} &= \mathbb{M}_{x}^{(4)} - \mathbb{M}_{x}^{\alpha(2)} - \mathbb{M}_{x}^{\beta(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_{x}^{(4)} - \mathbb{M}_{x}^{\alpha(2)} + \mathbb{M}_{x}^{\beta(2)} + \mathbb{T}_{yz}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 \\
0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} \\
T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0
\end{bmatrix}
$$

$$T_{yzxx} = \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)}$$
$$T_{yzyy} = \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)}$$
$$T_{yzzz} = \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)}$$
$$T_{yzzx} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{zxyz} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - \mathbb{T}_{xy}^{(6)}$$
$$T_{zxxy} = \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + \mathbb{T}_{yz}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyyy} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyzz} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$
$$T_{xyzx} = \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - \mathbb{T}_{yz}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & T_{xxxy} & T_{xxzy} & 0 & T_{xxyx} \\ 0 & 0 & 0 & T_{yyyz} & 0 & T_{yyxy} & T_{yyzy} & 0 & T_{yyyx} \\ 0 & 0 & 0 & T_{zzyz} & 0 & T_{zzxy} & T_{zzzy} & 0 & T_{zzyx} \\ T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} & T_{zxzy} & 0 & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$
$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxzy} = -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$
$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyyz} = 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$
$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyzy} = 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$
$$T_{yyyx} = -2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{zzyz} = -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$
$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{zzzy} = -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$
$$T_{zzyx} = -\mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{(4)} - 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{yzxx} = 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)}$$
$$T_{yzyy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$
$$T_{yzzz} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$
$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{zxyz} = \mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{zxxy} = -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$
$$T_{zxzy} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{zxyx} = -\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$
$$T_{xyxx} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyyy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyzz} = 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyzx} = \mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$
$$T_{xyxz} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} & T_{zxzy} & 0 & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\
T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{zxzy} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\
T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\
T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\
T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\
T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ T_{yyyx} & 0 & T_{yyyz} \\ 0 & T_{zzzy} & 0 \\ 0 & T_{yyzy} & 0 \\ 0 & T_{zzxy} & 0 \\ T_{xxyx} & 0 & T_{xxyz} \\ T_{yzzx} & 0 & T_{yzzz} \\ 0 & T_{zxxy} & 0 \\ 0 & T_{xyyy} & 0 \\ -\frac{T_{yyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxy} & 0 & \frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} - T_{yyyx} + T_{zzzy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxxy} &= \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyx} &= -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzxy} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{xxyx} &= \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} \\
T_{xxyz} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yzzx} &= 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} \\
T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxy} &= 2\mathbb{T}_{4x}^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyyy} &= -\mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}
\end{aligned}$$